

even if you know what services they're using. A really crass way to look at it is to imagine how much you'd care about your privacy settings on Facebook when you didn't have a toilet to use, and often, even doing your 'business' in the morning was a social affair that was done with others (of the same sex, obviously) from your village? Privacy is a luxury in most people's lives in India. Their real, everyday lives. For better or worse, this affects how they think about privacy online – and for the large part, they don't seem to care.

Social media profiles of women who live in very oppressed societies often reflect a level of escapism. While they may live veiled lives, on social media they're out and proud, confident in the fact that they're actually unrecognisable offline.

It's not just women, our males are not shy either, especially when it comes to approaching women online. Memes that stereotype us Indian males online as constantly seeking 'bobs' (and the other thing) didn't get started because of the odd request. Many white women are being flooded with such requests, because they're stereotyped as porn stars by uninformed Indian men. Shyness certainly doesn't seem to be a problem at all, and privacy is an afterthought.

MOVING

A large chunk of India is moving out of rural areas and sending kids to larger cities for education or work. This mass migration has been happening for years, and will continue unabated because everyone wants to earn more and have a better life. With more and more people moving to cities, it will mean more and more people also come online, because, when in Rome... when everyone around someone is online, the chances that they will get online soon are very, very high.

This may not be the best news for our already overcrowded cities, but it's going to happen nonetheless. There's another effect this has, because many

INTERESTING PROJECTS

RightMesh

(<https://dgit.in/RightMesh>)

What happens when you're a village without connectivity, or with only limited connectivity, can you still make use of smartphones? You can use smartphones like dumbphones so long as you have a 2G connection, but in some areas, even that is spotty, which makes your smartphone just feel like an expensive brick – you can't call, you can't text, and you certainly cannot use the internet. Not if RightMesh get their way! Using the blockchain service Ethereum, it assigns every phone a unique Ethereum ID, which is then used to identify it for life. Even when there's no network, there's always Bluetooth, Wi-Fi and Wi-Fi Direct, which allows phones to network with one another. So long as you're in range of other phones, you're on a network, and you can chat, or even browse offline web pages. If you make a large network and one node has a connection to the web, all of them can connect to the web as well. If you want to message an ID that's offline, the message circulates on the network (encrypted of course) and when the right node comes online, the message is delivered by whatever device the node connects to. And it doesn't stop at the local network (thought that's the bit we're interested in for this article), it can connect phones, IoT devices, web servers and more in a mesh network. It even connects mesh networks with other mesh networks, all depending on how much connectivity there is in the mesh. We can imagine entire villages using a technology like this to keep in touch and share information. Imagine one person streaming a match on his phone using his data connection, and sharing it with the whole village over Wi-Fi Direct. Copyright holders will throw a fit, but it sounds like just what the next billion might want. This is only available on Android devices.

of the people who move to cities are those who leave behind family in their villages. When they return on holiday, when they move back, or to get married, etc., they bring the experiences of city life with them. This results in more people being exposed to the Internet, and thus wanting to also be on it.

TECHNOLOGIES THAT WILL DRIVE CHANGE

So what are the technologies that will help get the Next 4 Billion online, and how will they do so? Obviously there's no one answer, and there will be some we can't even foresee, which will come up in the next decade, but we can certainly list some we know for sure will help.

OFFLINE INTERNET?

An oxymoron assuredly, but one that's needed in order to get people online. One of the biggest problems we might have is coverage. With almost all of the coverage provided by pri-

vate companies, it's no wonder that we have very large black spots all across India which get no coverage at all. People live there, but they're not numerous enough to merit the huge infrastructure investment that would be needed in order to connect them. Getting those people online will be the biggest challenge. Still, technology can play a role. One way to do that is to make sure that phones (which are going to be the only device most people in the world use to access the internet) are able to function and offer at least basic services out of the box, whether or not there is coverage, or even when there's just 2G coverage. A simple example is voice commands. Right now, you and I, on our expensive phones, cannot ask Google Assistant to do much when the phones are offline. The challenge is to get the cheapest of phones with really low processing power and often no net connection at all, to be able to converse with its owner, and also answer basic questions. Another example would be to pre-bundle

